

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent
Kind Code
Date of Patent
Inventor(s)

12407403
B2
September 02, 2025
Michel; François et al.

Method of allocation in an on-board data transmission network in a mobile passenger transport vehicle and associated computer program

Abstract

The present invention relates to an allocation method in a transmission network on-board a mobile vehicle, the network comprising wireless access points configured for transmitting and receiving, over a functioning channel, radio signals between said access point and a plurality of terminals. The method comprising the following steps: provision of a digital model of the mobile vehicle and of the access points; for every available channel and for every access point, simulation of the propagation of signals from said access point on said available channel; for every available channel, for every access point and for every terminal, determination of an estimated associated bandwidth according to the simulation; allocation of one of the available channels and one of the access points to every terminal according to the estimated bandwidths.

Inventors: Michel; François (Merignac, FR), Harambillet; Pierre (Merignac, FR), Boullet; Pierre (Merignac, FR)

Applicant: THALES (Courbevoie, FR)

Family ID: 79601866

Assignee: THALES (Courbevoie, FR)

Appl. No.: 17/973265

Filed: October 25, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230126393 A1	Apr. 27, 2023

Foreign Application Priority Data

FR	21 11406	Oct. 27, 2021
----	----------	---------------

Publication Classification

Int. Cl.: H04B7/185 (20060101); H04W16/22 (20090101); H04W24/02 (20090101);
H04W72/53 (20230101); H04W84/12 (20090101)

U.S. Cl.:

CPC H04B7/18506 (20130101); H04W16/225 (20130101); H04W24/02 (20130101);
H04W72/53 (20230101); H04W84/12 (20130101)

Field of Classification Search

CPC: H04B (7/18506); H04W (72/53); H04W (16/225); H04W (24/02); H04W (84/12)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6636737	12/2002	Hills et al.	N/A	N/A
2006/0280201	12/2005	Hammel et al.	N/A	N/A
2015/0103685	12/2014	Butchko	455/67.14	H04L 41/147
2023/0021462	12/2022	Valenza	N/A	H04B 17/3913

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
114448490	12/2021	CN	H04B 7/18502
1 182 897	12/2001	EP	N/A
WO-2005029278	12/2004	WO	H04W 16/00

OTHER PUBLICATIONS

French Search Report dated Jun. 1, 2022, for FR 2111406, 2 pp. cited by applicant

Primary Examiner: Avellino; Joseph E

Assistant Examiner: Emadi; Maryam

Attorney, Agent or Firm: NIXON & VANDERHYE

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims priority to FR 21 11406 filed Oct. 27, 2021, the entire contents of which are hereby incorporated by reference.

FIELD OF THE INVENTION

(2) The present invention relates to a method of allocation in an on-board data transmission network in a mobile vehicle for passenger transport.

(3) The invention further relates to a computer program associated with such an allocation method.

(4) The mobile vehicle is in particular, an aircraft suitable for carrying passengers in a cabin.

BACKGROUND

(5) Systems which allow passengers to connect from their cell phones, tablets or computers to an on-board Wi-Fi network are known, mainly on-board long-haul commercial aviation aircraft. Hereinafter, a Wi-Fi network is a network which complies with all IEEE 802.11 standards.

(6) The Wi-Fi network gives access to the Internet network via a satellite connection or via a ground-based antenna. Same is referred to as “IFC” (“In Flight Connectivity”).

(7) The Wi-Fi network also provides access to content (movies, music, games) stored in a server on-board the aircraft. Same refers to “Wireless IFE” (“Wireless In Flight Entertainment”).

(8) In a manner known per se, such an IFE system comprises a plurality of entertainment terminals, further called user terminals, generally in the form of individual screens or tablets. Every entertainment terminal is integrated e.g. into the passenger seat or into the seat in front of the passenger. Every entertainment terminal is, as a variant, a digital tablet or a smart phone.

(9) Thus arranged, the entertainment terminals allow passengers to consult multimedia contents during the flight (e.g. films, TV programs, games or music) and to be informed about the course of the flight (altitude, speed, current position, progress, etc.).

(10) In some cases, the entertainment terminals can be further used for giving certain practical information relating e.g. to the airport of arrival and for broadcasting announcements made by the crew in audio and/or video form.

(11) The Wi-Fi network consists of equipment, in particular WAP (Wireless Access Point) equipment. The terminal of every passenger is connected at a given time to a single WAP which provides the passenger with the data link. A WAP serves some of the passengers in the cabin. The number of WAPs and the locations thereof depend on the type of aircraft: a single-aisle aircraft has 3 WAPs, or 4 WAPs, and in general, a long-haul double-aisle aircraft has 6 WAPs. WAPs are typically installed between the cabin ceiling and the aircraft skin, generally close to the aircraft centerline.

(12) However, the metallic cabin environment and connection density (one user terminal per seat) severely limits the bandwidth available for every passenger equipment item. Furthermore, a communication between two user terminals necessarily goes through a WAP device.

(13) And yet, the Wi-Fi network was not designed for a dense environment, in other words a large number of customers in a small space, which is the case of an aircraft passenger cabin. In a cabin, in order to avoid interference, only one channel can be used at a given time between a WAP and a Wi-Fi customer, i.e. two WAPs cannot use the same channel at a given time.

(14) Moreover, the latest Wi-Fi standards use high frequency carriers (between 5 GHz and 7 GHz) which are sensitive to the masking by seats and luggage compartments. Thus, the reception of the Wi-Fi signal is variably good depending on the respective position of the WAP and of the Wi-Fi customer (tablet, phone, computers, etc.). Thus, there is a large disparity in the speeds accessible depending on the seat where the passenger is located.

(15) The two phenomena of interference and masking make it difficult to distribute a satisfactory speed to a whole cabin, e.g. an average speed greater than 2 Mbit/s per passenger.

(16) Thus, given such constraints, the number of users of a Wi-Fi network on-board an aircraft is limited. In fact, connectivity (IFC) offers for passengers are often paying, which greatly limits the number of users. There are, however, free connectivity offers or wireless IFE. In such case, the number of users is larger, but remains limited. Moreover, the uses are often divided between Internet browsing and messaging with low bandwidth consumption (<500 Kbit/s) and streaming which draws consumption (>2 Mbit/s). If the demand becomes too high on the Wi-Fi network, the bandwidth available for every passenger decreases resulting in a decrease in the quality of the passenger's experience with e.g. a low-resolution transition for streaming and a slowdown in Internet browsing.

(17) In order to assign the different access points and channels available to the different Wi-Fi

customers, the allocation of channels and WAP to the customers is usually static. Reallocations are made if a Wi-Fi customer breaks the link thereof to the WAP and requests to re-link. It is thus possible that the same channel is used by a large number of users while another channel is not saturated. Moreover, a user can be connected to a WAP located at the other end of the aircraft and hence with a deeply masked signal while the user could be connected to an unmasked WAP closer to the user.

(18) To try to overcome such problem, Wi-Fi communication systems implementing dynamic allocation management are known, e.g. by measuring the speed at every WAP in operation and proposing regular reallocation of customers according to such measurements. Such a dynamic allocation of channels and WAP improves the availability of the Wi-Fi network for users but does not give full satisfaction because the allocation does not take into account the particularities of the environment of every cabin, in particular the configuration of the aircraft, the masking areas and the location of the different customers. Moreover, such an allocation method does not guarantee a minimum bandwidth for passengers. Finally, such a method does not provide a differentiated policy amongst customers e.g. by ensuring a minimum speed for business class passengers on a commercial aircraft.

(19) There is thus a need to obtain an allocation method which can address the above-listed issues, in particular providing a better allocation of the different channels and available wireless access points, taking into account the particularities of the mobile vehicle.

SUMMARY

(20) To this end, the invention relates to an allocation method in a data transmission network on-board a mobile vehicle for passenger transport, the mobile vehicle preferentially being an aircraft, the transmission network comprising a plurality of wireless access points, each access point being configured for transmitting and receiving, on one functioning channel, a plurality of available channels, radio signals between said access point and a plurality of terminals on-board the mobile vehicle, the allocation method comprising the following steps: provision of a digital model of the mobile vehicle and of the access points; for each available channel and for each access point, simulation with the digital model of the propagation of radio signals in the mobile vehicle from said access point on said available channel; for each available channel, for each access point and for each terminal, determination of an estimated bandwidth associated with the communication between said access point and said terminal on said channel according to the simulation associated with said access point and said channel; allocation of one of the available channels and one of the access points to each terminal according to the bandwidths estimated using an optimization algorithm.

(21) According to other advantageous aspects of the invention, the allocation method comprises one or a plurality of the following features, taken individually or according to all technically possible combinations: each wireless access point is configured for communicating with devices according to the Wi-Fi protocol. each terminal is chosen from the group consisting of: a fixed entertainment terminal of the mobile vehicle and a portable electronic device; the communication network comprises at least one computer system chosen from the group consisting of: a communication system with an antenna arranged outside the mobile vehicle and a multimedia server on-board the mobile vehicle; the mobile vehicle comprises a cabin suitable for accommodating passengers, the numerical model comprising: a three-dimensional modeling of the cabin layout and the materials used in the cabin and a wave diagram associated with each access point and with each terminal; the optimization algorithm comprises the application of a degradation coefficient to each estimated bandwidth, the degradation coefficient being a function of the number of terminals connected to the access point associated with said estimated bandwidth; the optimization algorithm applies an optimization criterion chosen from the group consisting of: maximizing the average bandwidth for all terminals; maximizing the average bandwidth for all terminals assuming one of the access points is down; guaranteeing a minimum bandwidth for a

predetermined subset of terminals and then maximizing the average bandwidth for all terminals; maximizing the average bandwidth for all terminals with a predetermined maximum standard deviation; the simulation step is carried out: on a computer placed outside the mobile vehicle, all the estimated bandwidths being stored on a memory on-board the mobile vehicle; and/or on a computer on-board the mobile vehicle; the optimization algorithm is executed by means of a machine learning method trained on a database comprising the estimated bandwidths and advantageously the bandwidths measured in the mobile vehicle.

(22) The invention further relates to a computer program comprising software instructions which, when executed by a computer, implement a method as defined above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The features and advantages of the invention will appear upon reading the following description, given only as an example, and making in reference to the enclosed drawings, wherein:
- (2) FIG. 1 is a schematic representation of a mobile vehicle comprising a data transmission network;
- (3) FIG. 2 is a schematic representation of the cabin of the mobile vehicle shown in FIG. 1, and of the transmission network;
- (4) FIG. 3 is a graph showing the degradation coefficient of a bandwidth as a function of the number of connected terminals; and
- (5) FIG. 4 is an organization chart of an allocation method according to the invention.

DETAILED DESCRIPTION

- (6) A mobile vehicle **10** is shown in FIG. 1.
- (7) The mobile craft **10** is configured to carry passengers, in particular a few dozen passengers, or even a few hundred passengers.
- (8) The mobile vehicle **10** is herein an aircraft, in particular a commercial aviation aircraft, such as a long-haul aircraft.
- (9) The present invention is described below in the particular case of an aircraft. However, a person skilled in the art would understand that the present invention applies to any means of transporting passengers such as a railway vehicle, a boat, a motor vehicle, a bus, a submarine, etc.
- (10) As can be seen in FIGS. 1 and 2, the mobile vehicle **10** comprises a cabin **12** wherein are arranged a plurality of seats **14** for passengers **16**.
- (11) The mobile vehicle **10** further comprises a data transmission network **18**.
- (12) The transmission network **18** comprises a communication system **20** with at least one antenna **22** arranged outside the mobile vehicle **10** and/or a multimedia server **24** on-board the mobile vehicle **10**.
- (13) As can be seen in FIG. 1, the antenna **22** is e.g. a satellite or a ground-based antenna. The communication system **20** is suitable for sending or receiving data from the antennas **22** and in particular can be used for connecting the transmission network **18** to the Internet network. The communication system **20** provides an IFC (In Flight Connectivity).
- (14) The multimedia server **24** is configured for storing entertainment content and for communicating the entertainment content to the passengers **16** via the network **18**. The multimedia server **24** comprises e.g. films, TV programs, games or music, and/or information on the course of the flight (altitude, speed, current position, progress, etc).
- (15) As can be seen in FIG. 2, the network **18** comprises a controller **15** and a plurality of wireless access points **26**. The access points **26** are arranged in the cabin **12** of the mobile vehicle **10** and are advantageously distributed along the mobile vehicle **10**.
- (16) A plurality of terminals **30** are on-board in the mobile vehicle **10**.

- (17) Each terminal **30** is in particular a fixed entertainment terminal **32** of the mobile vehicle **10** or a portable electronic device **34**.
- (18) Each access point **26** is configured for transmitting and receiving, on a functioning channel assigned by the controller **15** among a plurality of available channels, radio signals between said access point **26** and the terminals **30**.
- (19) In particular, each access point **26** is suitable for functioning on a functioning channel of the allocated frequency band. The functioning channel is adjustable and assigned dynamically by the controller **15** to the access point **26**.
- (20) Advantageously, each wireless access point **26** is configured for communicating with the terminals **30** according to the Wi-Fi protocol, conforming to at least one of the versions of the IEEE 802.11 standard. The frequency of the carrier of the waves exchanged according to the Wi-Fi protocol is advantageously comprised between 5 GHz and 7 GHz.
- (21) Thus, each access point **26** is suitable for establishing a temporary Wi-Fi link with one of the terminals **30**, when this terminal **30** is located inside a coverage cell associated with the access point **26** in question.
- (22) The coverage cells of the different access points **26** overlap, at least two by two, so as to cover, bit by bit, most of or even the entire cabin **12**.
- (23) As shown in FIG. 2, each entertainment terminal **32** is known per se and comprises in particular, a display screen, preferentially a touch screen. Each entertainment terminal **32** is e.g. fixed or integrated into the seat **14** of the passenger **16**, or is fixed or integrated into the backrest of the seat **14** located in front of the seat **14** of the passenger **16**.
- (24) Each entertainment terminal **32** is configured for delivering to the passengers **16** the content stored in the multimedia server **24**, in particular, following a selection by a passenger **16** of a field displayed on the screen of entertainment terminal **32** thereof.
- (25) Each portable electronic device **34** is typically a Personal Electronic Device (PED) belonging to a passenger **16**. Each portable electronic device **34** is e.g. a smartphone or an electronic tablet.
- (26) The controller **15** comprises a simulation module **40** and an allocation module **42**.
- (27) The controller **15** further comprises a database **44**.
- (28) The database **44** is configured for storing in memory a digital model of the mobile vehicle **10** and of the access points **26**.
- (29) The numerical model comprises a three-dimensional modeling of the layout of the cabin **12**.
- (30) The three-dimensional model comprises in particular, a representation of the external and internal walls of the mobile vehicle **10** and in particular of the cabin **12**, of the seats **14** arranged in the cabin **12** and of the possible masking zones of the access points **26**.
- (31) The digital model further comprises the materials used in the cabin **12**, and in particular the conductivity thereof and the permeability thereof to radio waves. As an example, a metal wall significantly disturbs the transmission of radio waves. On the other hand, glass provides a relative good transmission for radio waves.
- (32) The digital model further comprises a wave diagram associated with each access point **26** and with each terminal **30**. A wave diagram is a spatial representation of the angular radiation distribution of the access point **26** or of the terminal **30**.
- (33) The simulation module **40** is suitable for, for each available channel and for each access point **26**, simulating with the digital model, the propagation of radio signals from said access point **26** to said computer terminal **30** on said available channel.
- (34) The simulation module **40** is further suitable for determining, for each available channel, for each access point **26** and for each terminal **30**, an estimated bandwidth associated with the communication between said access point **26** and said terminal **30** on said channel.
- (35) The bandwidth indicates the capacity of the network to transmit information. The bandwidth represents the volume of digital data which can be transferred between said access point **26** and said terminal **30** on said channel in a given time interval.

(36) In an alternative embodiment, the simulation module **40** is arranged in a computer (not shown) outside the mobile vehicle **10**. When the mobile vehicle **10** is an aircraft, the computer is in particular located on the ground.

(37) The computer comprises another database suitable for storing the digital model. The database **44** arranged in the mobile vehicle **10** then does not comprise the digital model. The database **44** is suitable for storing the result of the simulations carried out by the simulation module **40** on the ground, in particular the estimated bandwidth.

(38) The allocation module **42** is configured for allocating one of the available channels and one of the access points **26** to each terminal **30** according to the bandwidths which were estimated by an optimization algorithm. Said access points **26** and said terminal **30** are then suitable for communicating and exchanging data on said allocated channel.

(39) The functioning of the simulation module **40** and of the allocation module **42** will be explained in more detail thereafter.

(40) In the example shown in FIG. 2, the controller **15** comprises an information processing unit consisting e.g. of a memory and of a processor associated with the memory. The simulation module and the speech module consist each of a software, or a software brick, executable by the processor. The memory is then apt to store a receiver software, a speech-recognition software, a processing software, a surveillance software, and, as an optional addition, a transmission software and a validation software. The processor is then apt to run each of said software.

(41) In a variant (not shown), the simulation and the speech module consist each of a programmable logic component, such as an FPGA (Field Programmable Gate Array), or further of a dedicated integrated circuit, such as an ASIC (Application Specific Integrated Circuit).

(42) When the controller **15** consists of one or a plurality of software programs, i.e. of a computer program, same is further apt for being recorded on a computer-readable medium (not shown). The computer-readable medium is e.g. a medium apt to store the electronic instructions and to be coupled to a bus of a computer system. As an example, the readable medium is an optical disk, a magneto-optical disk, a ROM memory, a RAM memory, any type of non-volatile memory (e.g. EPROM, EEPROM, FLASH, NVRAM), a magnetic card or an optical card. A computer program containing software instructions is then stored on the readable medium.

(43) An allocation method **100** according to the invention will be henceforth explained with the help of FIG. 4 showing a flowchart of the allocation method according to the invention.

(44) The method according to the invention will thereafter be described for the case of an aircraft, but a person skilled in the art would understand that such method applies in a similar manner to any mobile vehicle **10**.

(45) Initially, an aircraft including a cabin **12** for receiving passengers **16** is provided.

(46) A digital model of the aircraft and of the wireless access points **26** arranged along the cabin, during an initial stage **110** is also provided.

(47) As an example, in a single-aisle aircraft, the cabin **12** comprises 36 seats **14** in the so-called "Business" class and 156 seats **14** in the so-called "Economy" class. The transmission network **18** comprises 4 wireless access points **26** distributed along the cabin **12**.

(48) The allocation method **100** then comprises a simulation step **120**. In particular, for each available channel and for each access point **26**, a simulation **120** is carried out with the digital model of the propagation of radio signals in the aircraft **10**, and in particular in the cabin **12**, from said access point **26** onto said available channel.

(49) Due to the numerical model, such simulation takes into account the precise configuration of the aircraft, and in particular the geometry thereof and the materials forming same and influencing the propagation of waves in cabin **12**.

(50) The simulation makes it possible to obtain a mapping of the power available at every point of the cabin **12**, for each access point **26** and for each available channel.

(51) Then, during a step **130**, for each available channel, for each access point **26** and for each

terminal **30**, the method comprises a step of determining an estimated bandwidth associated with the communication between said access point **26** and said terminal **30** on said channel.

(52) In particular, the power mapping is cross-checked with the applicable Wi-Fi standard, the intrinsic features of information encoding on the signal and in particular the choice of the modulation scheme, in order to deduce the theoretical bandwidth achievable between said access point **26** and said terminal **30** on said channel.

(53) In one embodiment, the simulation step **120** is performed on a computer located outside the aircraft. The simulation step is performed upstream of the aircraft flight. The computer is in particular located on the ground and makes it possible in particular to have a high computing power. Furthermore, it is possible to run the simulations several times by modifying the desired model or optimization criteria.

(54) All the estimated bandwidths are then stored on a memory on-board the aircraft.

(55) As a variant, the simulation step **120** is carried out on a computer on-board the aircraft, in particular, during the flight of the aircraft. In such case, it is possible to carry out a more dynamic simulation taking into account the real situation on-board the aircraft, in particular the number of passengers **14** and of terminals **30** present in the cabin **12**.

(56) The method then comprises a step **140** of allocation of one of the available channels and of one of the access points **26** to each terminal **30** according to the bandwidths which were estimated by an optimization algorithm.

(57) The optimization algorithm includes in particular, the application of a degradation coefficient to each estimated bandwidth. The degradation coefficient is a function of the number of terminals **30** connected to the access point **26** associated with said estimated bandwidth. Such degradation coefficient reflects the behavior of terminals **30** of the access point **26** as a function of density. The degradation coefficient is directly related to the intrinsic technical characteristics of the access point **26**.

(58) As an example, FIG. **3** shows the degradation coefficient as a function of the number of terminals **30** connected. When no terminal **30** is connected, the coefficient is equal to 100% and thus no degradation is applied to the estimated bandwidth. When more than 80 terminals **30** are connected, a coefficient of 90% is applied to the estimated bandwidth. When more than 120 terminals **30** are connected, a coefficient of 50% is applied to the estimated bandwidth.

(59) The optimization algorithm allows the channels and access points **26** to be allocated to the different terminals **30** by applying an optimization criterion.

(60) In one variant, the optimization criterion aims to maximize the average bandwidth for all terminals **30**. In other words, the optimization criterion aims to ensure that, on average, each terminal **30** has as much available bandwidth as possible.

(61) In a variant, the optimization criterion aims to maximize the average bandwidth for all the terminals **30**, assuming that one of the access points **26** is down. In other words, the optimization criterion then makes it possible to take into account, a possible failure of an access point and makes it possible to determine an allocation for limiting the degradation of the service for the terminals **30**.

(62) In a variant or in addition, the optimization criterion aims to guarantee a minimum bandwidth for a predetermined subset of terminals **30** and then to maximize the average bandwidth for all terminals **30**. The predetermined subset of terminals **30** is e.g. the “Business” class. The optimization criterion makes it possible e.g. to ensure a minimum of 2 Mbit/s for each terminal **30** present in the “Business” class. Then, the optimization criterion aims to ensure that all terminals **30** have the largest available bandwidth, with the constraint relating to the “Business” class. As another example, in the event of a failure of one of the access points **26**, in order to limit the degradation of the service for the “Business” class, the optimization criterion makes it possible to ensure a minimum of 500 Kbit/s for each terminal **30** present in the “Business” class.

(63) In a variant or in addition, the optimization criterion aims to maximize the average bandwidth

for all terminals **30** with a predetermined maximum standard deviation. In other words, the optimization criterion aims at an equal allocation of channels and access points, with a predetermined maximum allowed standard deviation. The lower the maximum allowed standard deviation, the more equal the bandwidth distribution will be between the different terminals.

(64) In an advantageous embodiment, the allocation is made as a function of the level of filling of the aircraft, in particular of the number of terminals **30** present in the aircraft.

(65) Indeed, filling rates fluctuate according to flights. In order to optimize performance, configurations are calculated according to the different filling levels of the “Business” class and of the “Economy” class. Configuration means a set of assignments between the access points **26**, the channels, and the terminals **30**.

(66) As shown in Table 1 below, optimal configurations are calculated based on the number of passengers **16** in each class. Herein, fifteen configurations are pre-calculated upstream of the flight. Depending on the number of passengers **16** connected in each class, the controller **15** selects and applies the configuration pre-calculated during step **130**.

(67) E.g. if during the flight, fifteen passengers of the “Business” class and one hundred and five passengers **16** of the “Economy” class connect, then the controller **15** applies the configuration Conf-9. Thus, the access points **26** and the channels are assigned to the different terminals **30** according to said configuration.

(68) TABLE-US-00001

TABLE 1	Number of passengers in Economy Class	0-30	30-60	60-90	90-120	120-156	Number of	0-10	Conf-1	Conf-2	Conf-3	Conf-4	Conf-5	passengers	10-20	Conf-6	Conf-7	Conf-8	Conf-9	Conf-10	in Business	20-36	Conf-11	Conf-12	Conf-13	Conf-14	Conf-15
Class																											

(69) Advantageously, all the pre-calculated configurations (Conf-1; . . . ; Conf-15) observe the minimum speed of 2 Mbit/s for each terminal of the “Business” class which represented a limit case since the configuration was calculated for a full aircraft. Such configurations make it possible to obtain more resources (i.e. speed) for each terminal than in the case of a full aircraft.

(70) In addition (not shown), the table is completed in order to predict the failure of one of the access points **26** by adding to the above table, a dimension representing the “no failure” case and the cases of failure of each one of the access points **26**. In the case of a single-aisle aircraft, the above situation would form a set of 75 configurations to be applied according to the number of passengers **16** in “Economy” class, the number of passengers **16** in “Business” class and the possible failure of one of the access points **26**.

(71) The allocation algorithm is advantageously implemented by means of a combinatorial optimization algorithm, e.g. by means of the so-called greedy algorithm.

(72) Alternatively, in an advantageous embodiment, the optimization algorithm is executed by means of a machine learning method trained on a database comprising the estimated bandwidths. The machine learning method is e.g. a neural network or another equivalent model. The machine learning method is trained on the basis of the simulations carried out during the step **120**.

Advantageously, the model obtained is on-board the mobile vehicle **10** and continues to learn with measurements made under dynamic conditions in the cabin **12** during the flight.

(73) Such model takes as input, the data (e.g. the number of passengers) which will influence the maximum speeds and supplies as output, a table of the maximum speeds per terminal **30**.

(74) Initially, the model is configured for providing exactly the same results as the simulation carried out in the step **120** with the numerical model of the cabin **12**. Then, depending on the maximum speeds observed during actual flights, the model is adapted to best reflect the maximum achievable speeds.

(75) E.g. depending on the number of passengers **16** and the duration of the flight, the luggage compartments are filled to a greater or lesser extent, which modifies the propagation of Wi-Fi waves and hence influences the maximum achievable speeds. Such an approach makes it possible to adapt the maximum speed tables and thus the allocation of the access points **26** and of the

channels according to the conditions of the flight.

(76) In an advantageous embodiment, the allocation step **140** is carried out dynamically during the flight of the aircraft in order to adapt to the real situation of the cabin **12**.

(77) The allocation step **140** is then implemented in the aircraft. The controller **15** dynamically adapts the configuration of the channels and of the access points **26** according to the actual connections of the terminals **30** in the cabin **12**. Thus, during a flight of the aircraft, according to the changes in the connections, the algorithm recalculates a new configuration which is applied and replaces the old configuration.

(78) In particular, the triggering of the new iteration of the step of determination of a new configuration is carried out periodically, e.g. each ten minutes.

(79) In a variant, the triggering of the new iteration is carried out following a significant change in the number of connections. E.g. a first configuration is calculated with fifty terminals **30** connected. As long as the number of connections remains within the interval between forty and sixty connections, the controller **15** does not launch a new iteration of the allocation step **140**. When the number of connections falls below forty or raises above sixty, the controller **15** launches an allocation step **140** in order to obtain a new allocation more suitable for the current situation in the cabin **12**.

(80) In a variant, the triggering of the new iteration is carried out following a change in the quality of the connections of the terminals **30**. In particular, a monitoring of the quality of the connections is carried out and sent back to the controller **15**. When the quality of the connections deteriorates, in particular when the average speed per terminal **30** falls below a predetermined threshold or when a predetermined number of terminals has a speed below a threshold, the controller launches a new iteration of the allocation step **140**.

(81) Following the allocation, the passengers **16** have access to the IFC or IFE resources via the associated terminal **30** thereof which communicates with the access point **26** allocated on the allocated channel.

(82) It can be understood that the invention has a certain number of advantages.

(83) The allocation method according to the invention can be used for a better allocation of the different wireless channels and access points **26** available in the cabin **12**, taking into account the particularities of the mobile vehicle **10**.

(84) Simulation by means of a numerical model makes it possible to take into account the geometry of each cabin **12**, the location of the different access points **26** and terminals **30** in the cabin **12**, the masking zones and the materials used which can disturb the propagation of the waves to the various terminals **30**.

(85) The invention can then be used for efficiently allocating the access points **26** and the channels according to the optimization criterion chosen for the allocation.

(86) The invention can be further used for obtaining an allocation of Wi-Fi resources according to a minimum level of service defined a priori according to the category of passengers. It is in particular [possible] to guarantee a minimum bandwidth a priori, which is impossible with conventional methods. The invention is thus particularly advantageous when the IFC or IFE supplier commits to a quality experience for every passenger. In particular, the invention makes it possible to allocate additional resources to a particular category (e.g. the “Business” class) in order to guarantee a higher-quality service for said category of passengers.

(87) The invention further makes it possible to define a priori, a minimum service in the event of a failure of one of the access points **26**, and in particular to limit the impact on a certain category of passengers.

Claims

1. An allocation method in a data transmission network on-board a mobile vehicle for transporting passengers, the transmission network comprising a plurality of wireless access points, each access point being configured for transmitting and receiving, on one functioning channel among a plurality of available channels, radio signals between said access point and a plurality of terminals on-board the mobile vehicle, the allocation method comprising the following steps: provision of a digital model of the mobile vehicle and of the access points; for every available channel and for every access point, simulation with the digital model of the propagation of radio signals in the mobile vehicle from said access point on said available channel; for each available channel, for each access point and for each terminal, determination of an estimated bandwidth associated with the communication between said access point and said terminal on said channel according to the simulation associated with said access point and said channel; allocation of one of the available channels and one of the access points to each terminal according to the bandwidths which were estimated by an optimization algorithm.
 2. The allocation method according to claim 1, wherein the mobile vehicle is an aircraft.
 3. The allocation method according to claim 1, wherein each wireless access point is configured for communicating with the terminals according to the Wi-Fi protocol.
 4. The allocation method according to claim 1, wherein each terminal is chosen from the group consisting of: a fixed entertainment terminal of the mobile vehicle; and a portable electronic device.
 5. The allocation method according to claim 1, wherein the communication network comprises at least one computer system chosen from the group consisting of: a communication system with an antenna arranged outside the mobile vehicle; and a multimedia server on-board the mobile vehicle.
 6. The allocation method according to claim 1, wherein the mobile vehicle comprises a passenger cabin, the digital model comprising: a three-dimensional modeling of the layout of the cabin and the materials used in the cabin; and a wave diagram associated with each access point and with each terminal.
 7. The allocation method according to claim 1, wherein the optimization algorithm comprises the application of a degradation coefficient to each estimated bandwidth, the degradation coefficient being a function of the number of the terminals connected to the access point associated with said estimated bandwidth.
 8. The allocation method according to claim 1, wherein the optimization algorithm applies an optimization criterion chosen from the group consisting of: maximizing the average bandwidth for all terminals; maximizing the average bandwidth for all terminals assuming that one of the access points is down; guaranteeing a minimum bandwidth for a predetermined subset of terminals and then maximizing the average bandwidth for all terminals; maximizing the average bandwidth for all terminals with a predetermined maximum standard deviation.
 9. The allocation method according to claim 1, wherein the simulation step takes place: on a computer placed outside the mobile vehicle, all the estimated bandwidths being stored on a memory on-board the mobile vehicle; and/or on a computer on-board the mobile vehicle.
 10. The allocation method according to claim 1, wherein the optimization algorithm is executed by means of a machine learning method trained on a database comprising the estimated bandwidths.
 11. The allocation method according to claim 10, wherein the database in addition comprises bandwidths measured in the mobile vehicle.
 12. A non-transitory computer program comprising software instructions which, when being executed by a computer, implements an allocation method according to claim 1.
-